



PHENIKAA UNIVERSITY
Faculty of Computer Science

Software Analysis and Design

Module 4: Requirement Analysis with Use-Case Model

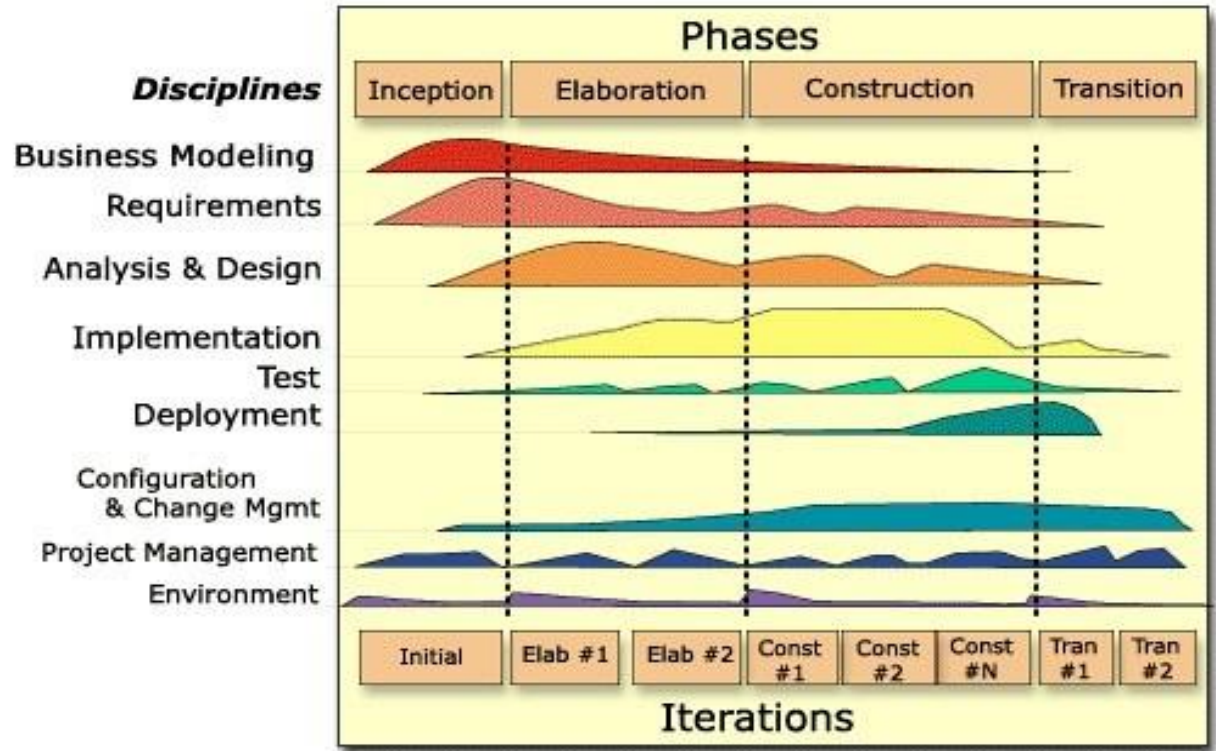
- ❖ Describe the basic Requirements concepts and how they affect Analysis and Design
- ❖ Demonstrate how to read and interpret the artifacts of Requirements that are used as a starting point for Analysis and Design

- ❖ Purpose of Requirement Analysis
- ❖ Use-Case Model
- ❖ Glossary
- ❖ Supplementary Specification

Purpose of Requirement Analysis



- ❖ Establish and maintain an agreement between customers and other stakeholders on what the system should do
 - ❖ Don't say how to do it
- ❖ Give system developers a better understanding of system requirements

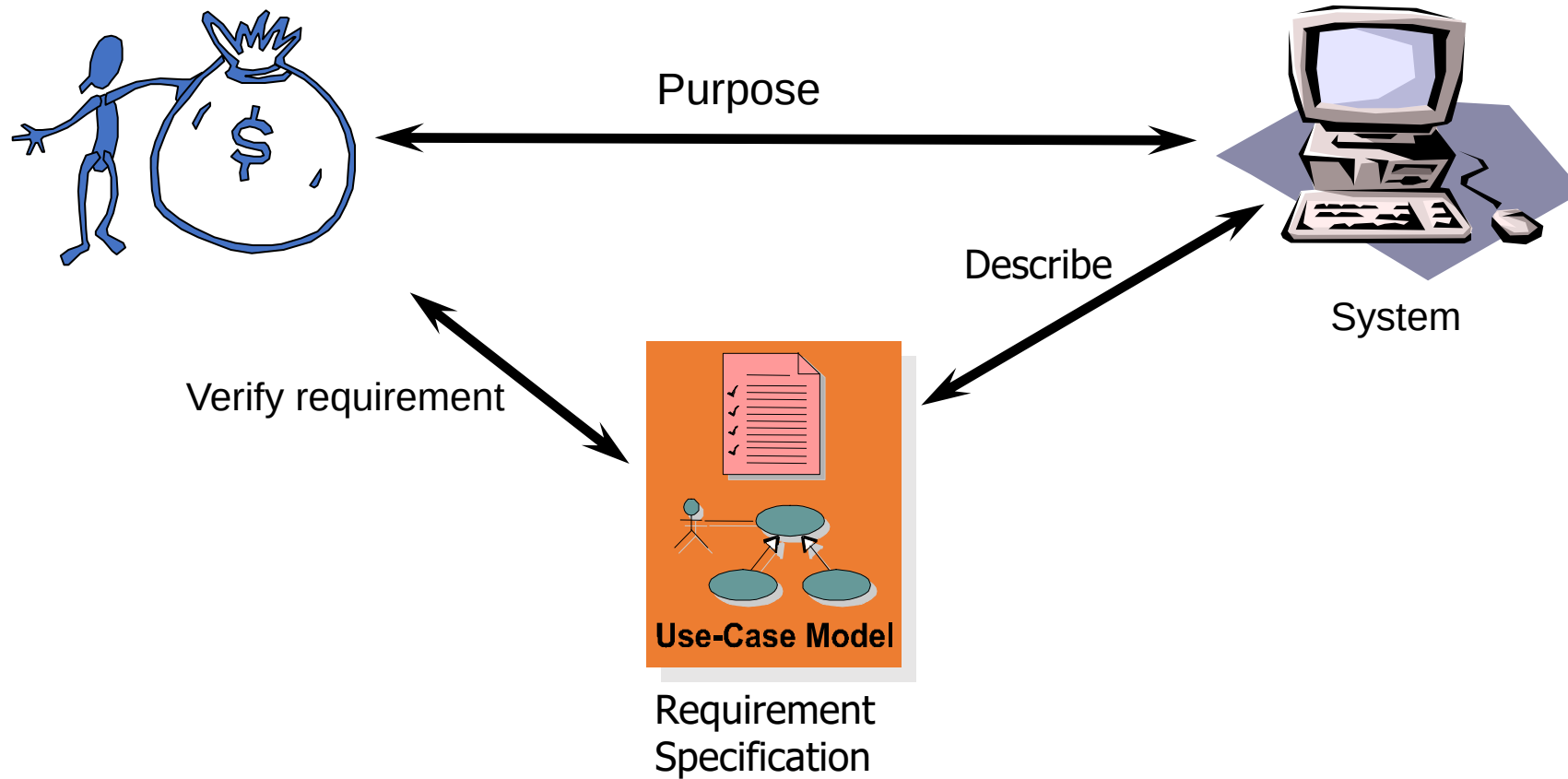


Purpose of Requirement Analysis



- ❖ Define what the system will and will NOT do
- ❖ Provide basic information for project development planning
- ❖ Provides the basis for estimating the cost and time to develop the system
- ❖ Identify the user interface of the system, capture the requirements and purposes of the user
- ❖ As input to the next phase of the project: Analysis and Design

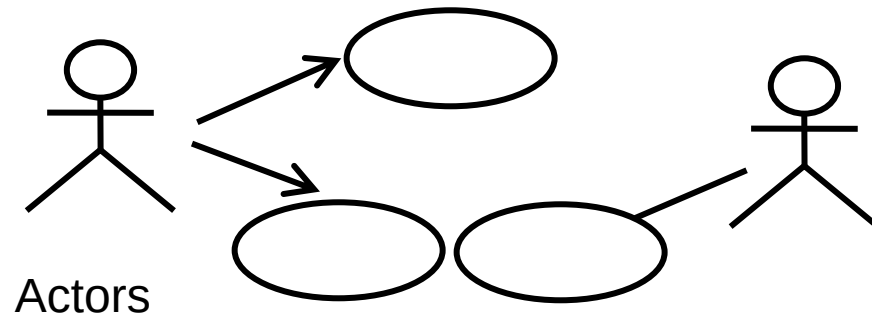
Purpose of Requirement Analysis



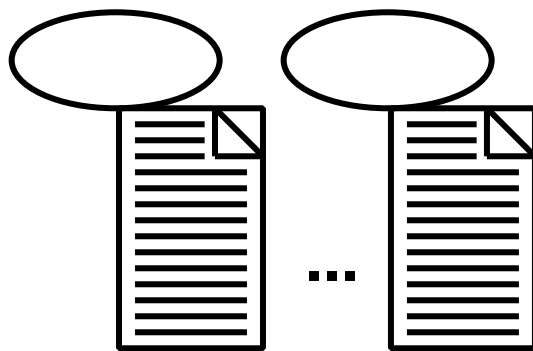
Relevant Requirements Artifacts



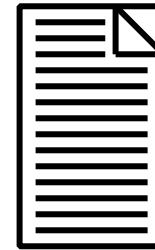
Use-Case Model



Use Cases



Use-Case Specifications



Glossary



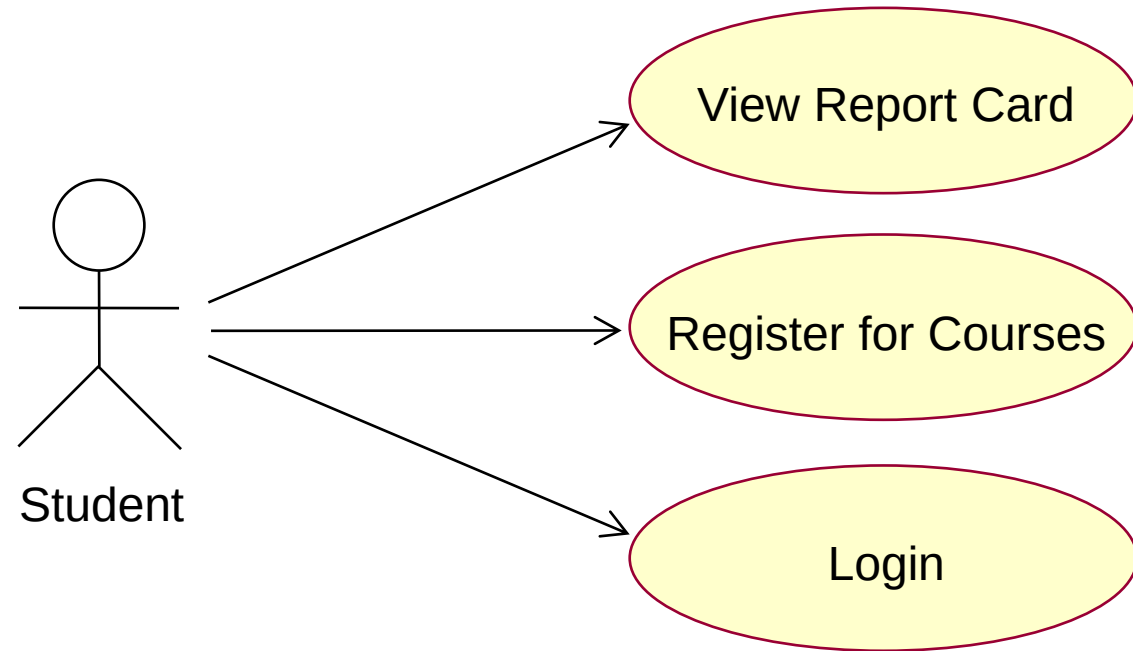
Supplementary
Specification

- ❖ Purpose of Requirement Analysis
- ❖ Use-Case Model
- ❖ Glossary
- ❖ Supplementary Specification

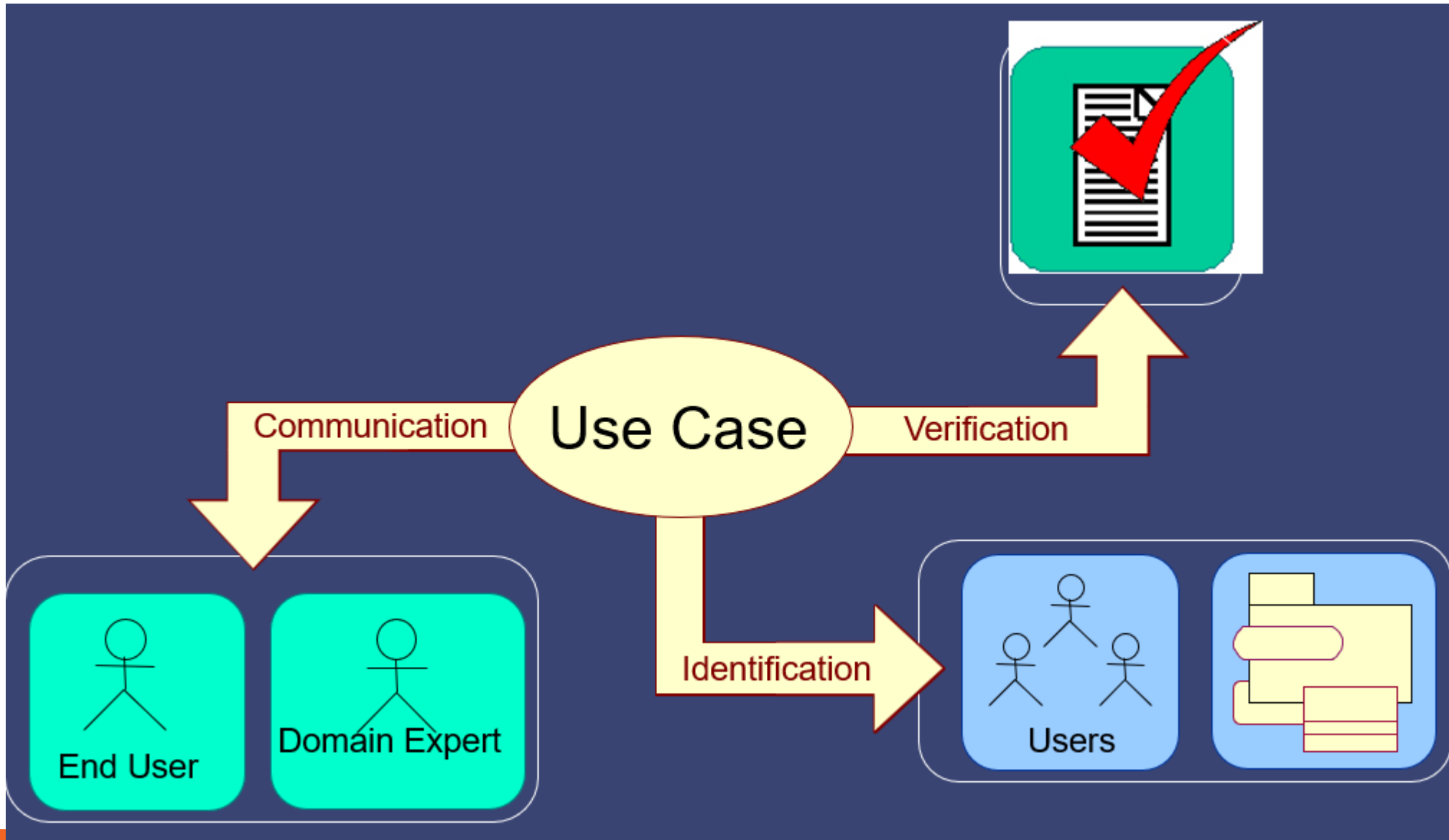
Use-case Model



- ❖ Modeling the functions that the system will perform
- ❖ The model includes the predetermined functions of the system
- ❖ Using the concept of Use Case



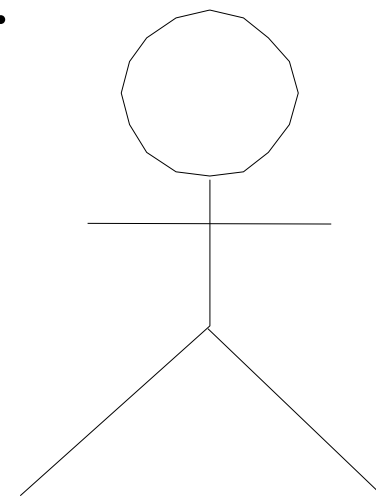
Benefits of Use-case Model



What is an actor



- ❖ Actors represent roles a user of the system can play.
- ❖ They can represent a human, a machine, or another system.
- ❖ They can actively interchange information with the system.
- ❖ They can be a giver of information.
- ❖ They can be a passive recipient of information.
- ❖ Actors are not part of the system.
 - ❖ Actors are EXTERNAL.

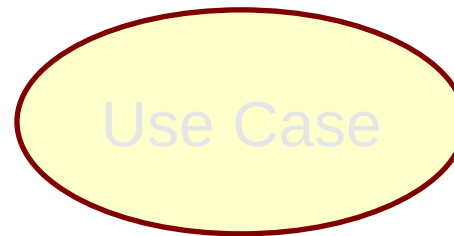


Actor

What is a use-case?



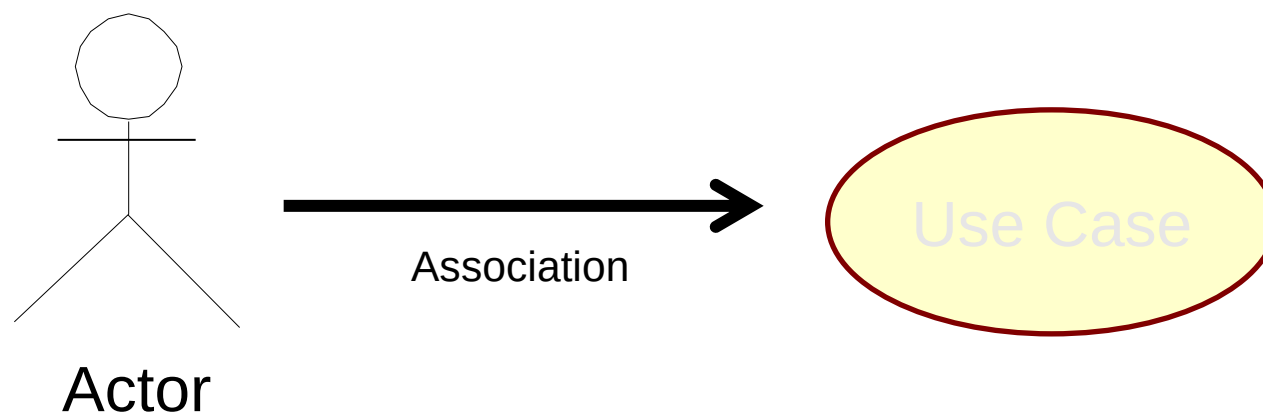
- ❖ Defines a set of use-case instances, where each instance is a sequence of actions a system performs that yields an observable result of value to a particular actor.
- ❖ A use case models a dialogue between one or more actors and the system
- ❖ A use case describes the actions the system takes to deliver something of value to the actor



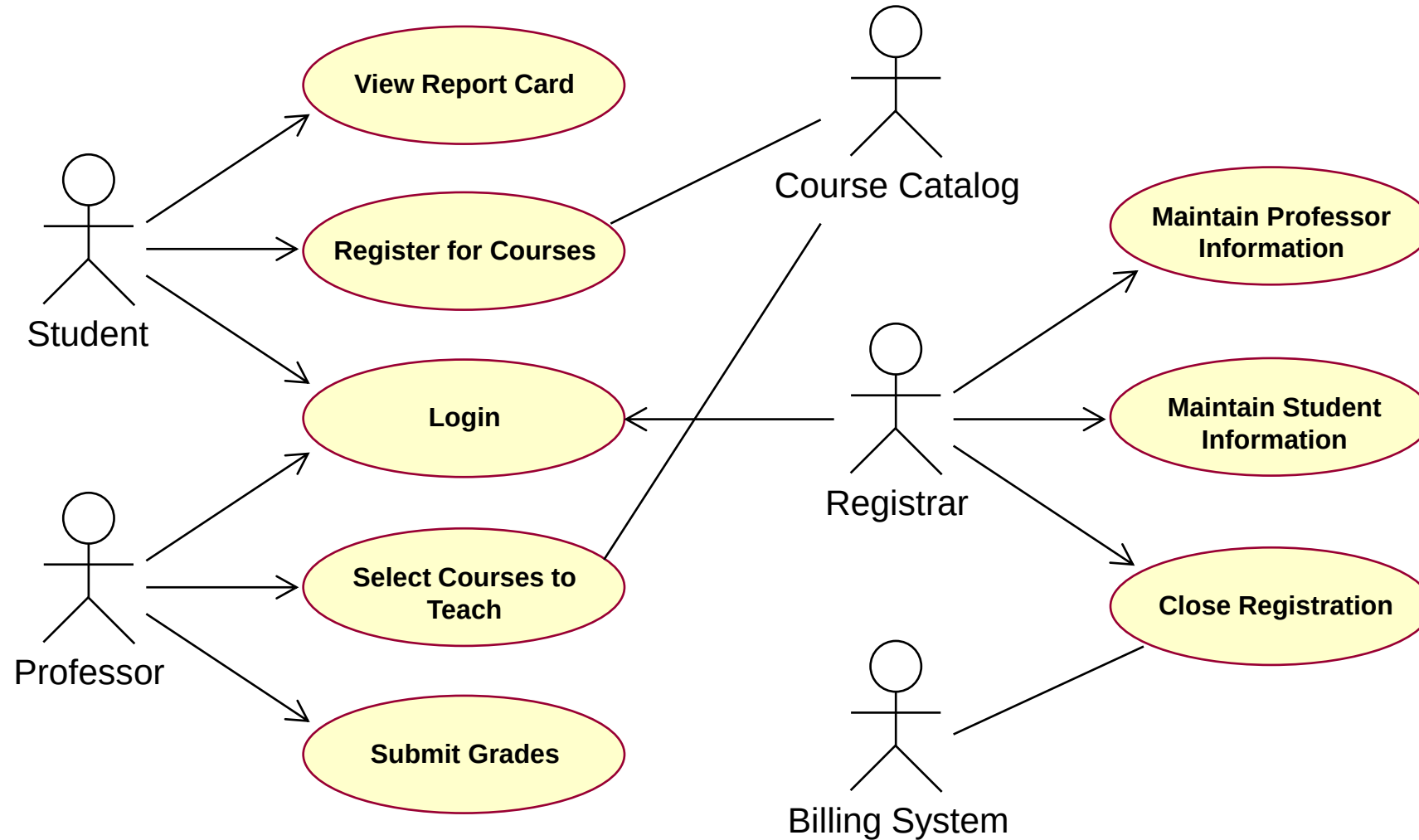
Use-case diagram



- ❖ A use case models a dialog between actors and the system.
- ❖ A use case is initiated by an actor to invoke a certain functionality in the system



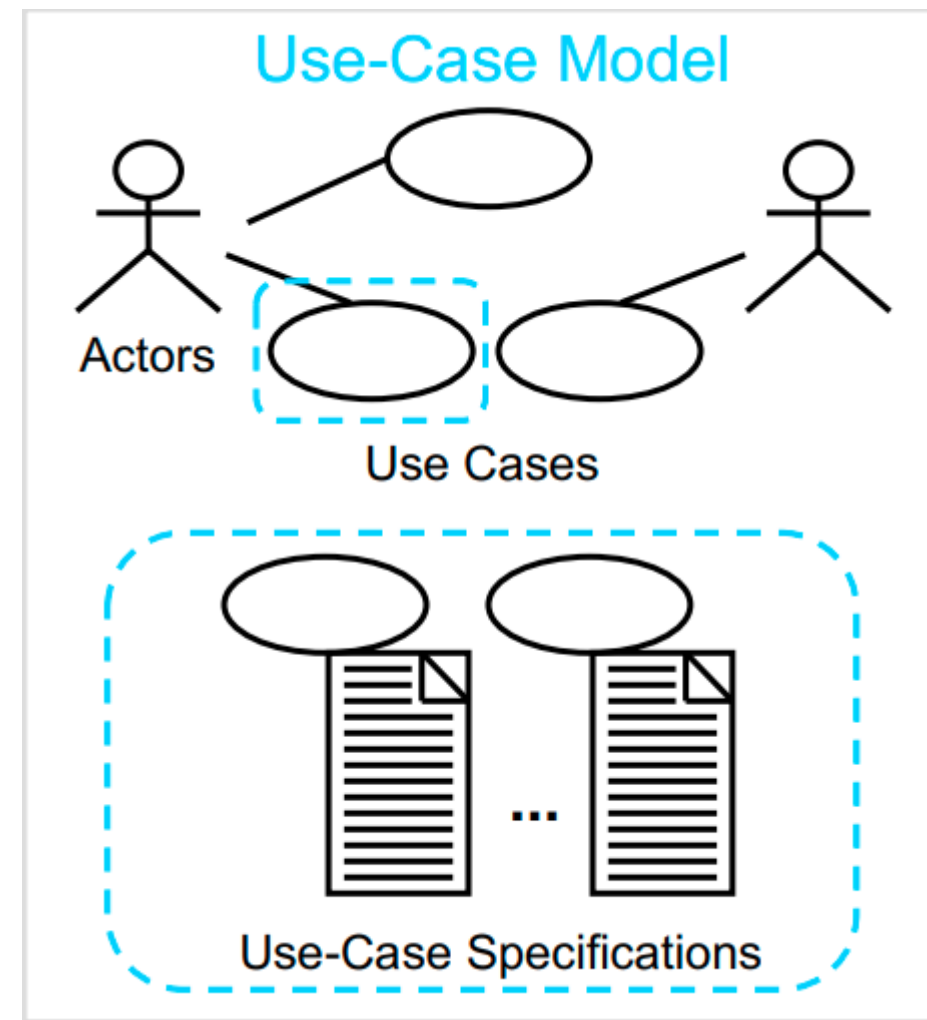
Use Case Diagram of Courses Registration System



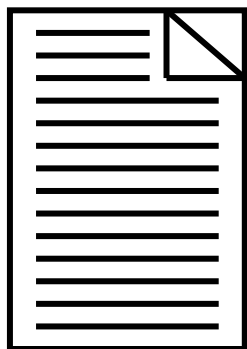
Use Case Specification



- ❖ Name
- ❖ Brief description
- ❖ Flow of Events
 - ❖ Basic flow
 - ❖ Alternative Flow
- ❖ Pre-conditions
- ❖ Post-conditions
- ❖ Extension points



- ❖ Purpose of Requirement Analysis
- ❖ Use-Case Model
- ❖ Glossary
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Course Registration System Glossary

1. Introduction

This document is used to define terminology specific to the problem domain, explaining terms, which may be unfamiliar to the reader of the use-case descriptions or other project documents. Often, this document can be used as an informal *data dictionary*, capturing data definitions so that use-case descriptions and other project documents can focus on what the system must do with the information.

2. Definitions

The glossary contains the working definitions for the key concepts in the Course Registration System.

2.1 Course: A class offered by the university.

2.2 Course Offering: A specific delivery of the course for a specific semester – you could run the same course in parallel sessions in the semester. Includes the days of the week and times it is offered.

2.3 Course Catalog: The unabridged catalog of all courses offered by the university.

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Supplementation Specification



- ❖ Functionality
- ❖ Usability
- ❖ Reliability
- ❖ Performance
- ❖ Supportability
- ❖ Design constraints

Functionality

Multiple users must be able to perform their work concurrently.

If a course offering becomes full while a student is building a schedule including that offering, the student must be notified.

Usability

The desktop user-interface shall be Windows 95/98 compliant.

Reliability

The system shall be available 24 hours a day 7 days a week, with no more than 10% down time.

Performance

The system shall support up to 2000 simultaneous users against the central database at any given time, and up to 500 simultaneous users against the local servers at any one time.

The system shall provide access to the legacy course catalog database with no more than a 10 second latency.

Note: Risk-based prototypes have found that the legacy course catalog database cannot meet our performance needs without some creative use of mid-tier processing power

The system must be able to complete 80% of all transactions within 2 minutes.

Supportability

None.

Security

The system must prevent students from changing any schedules other than their own, and professors from modifying assigned course offerings for other professors.

Only Professors can enter grades for students.

Only the Registrar is allowed to change any student information.



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Q&A